Assignment2

1)

코드

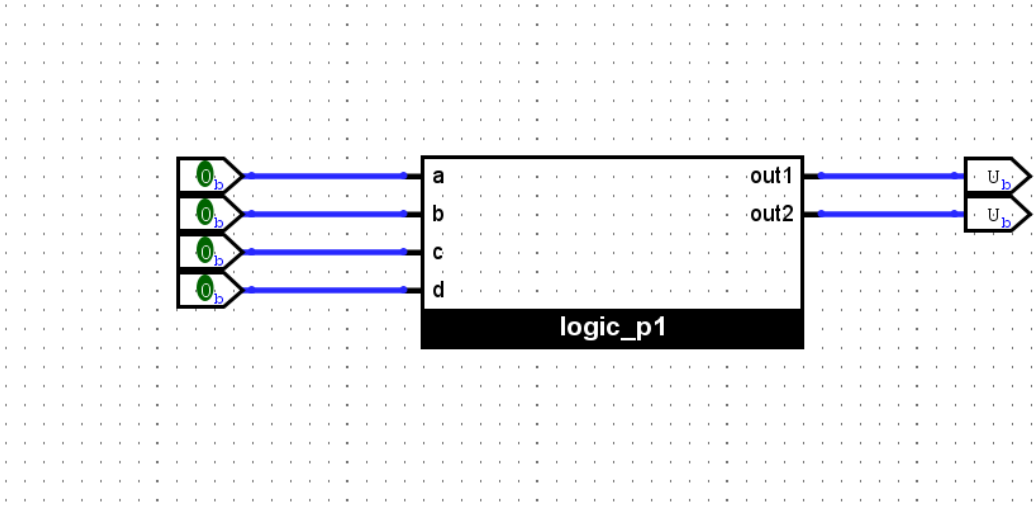
C: > Users > dlrjs > Desktop > 디논설 과제2 > problem_1.vhd

```vhdl
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

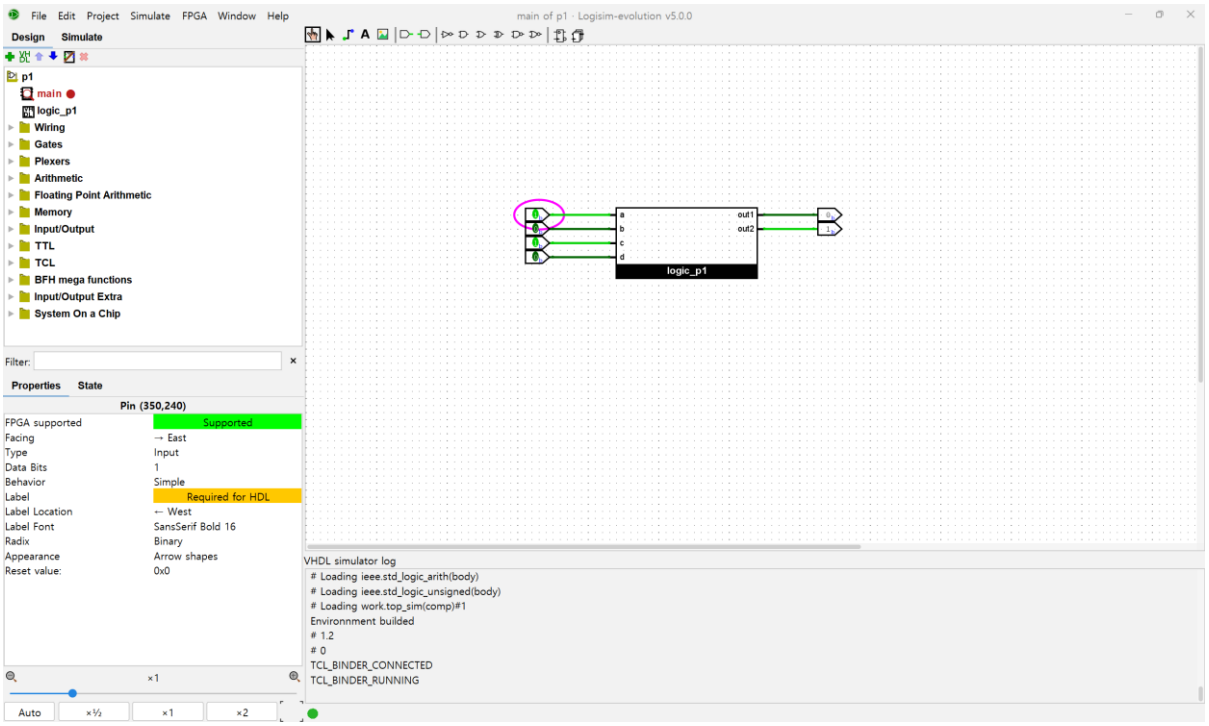
entity logic_p1 is
    port(
        a,b,c,d :in std_logic;
        out1,out2 :out std_logic
    );
end entity logic_p1;

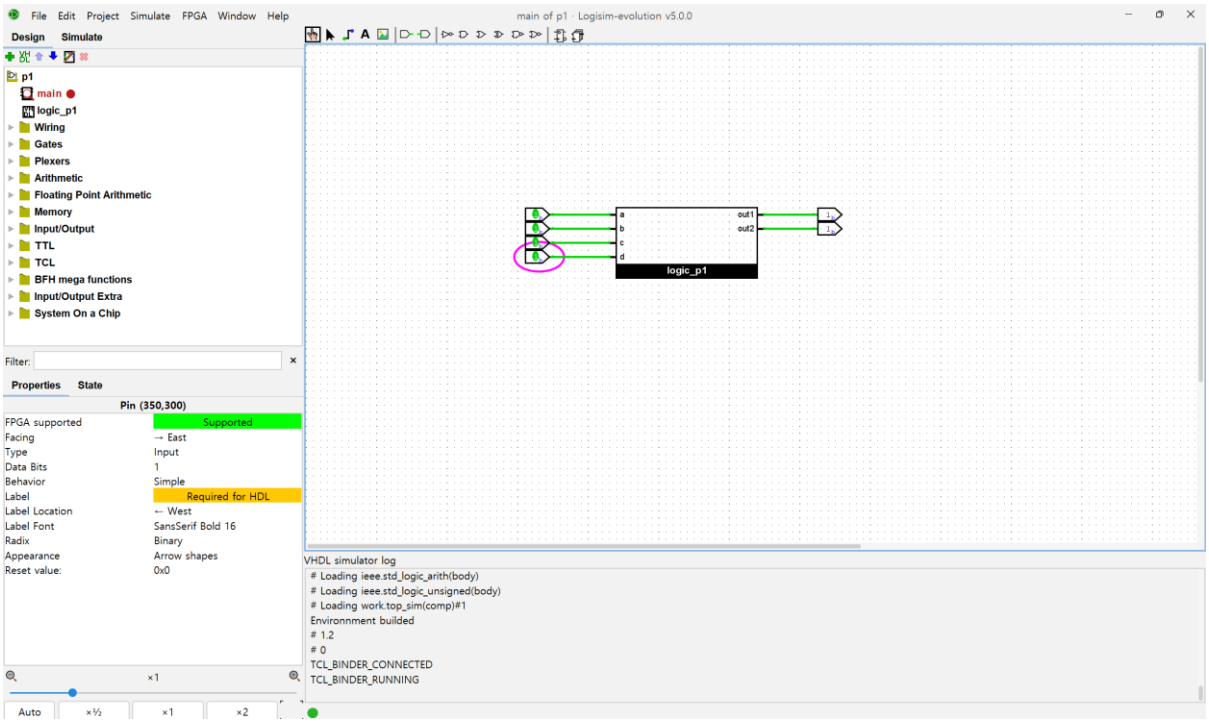
architecture bhv of logic_p1 is
begin
    out1 <= a and b and c and d;
    out2 <= a or b or c or d;
end architecture bhv;
```

회로



시뮬레이션 결과





AND, OR gate연산 회로이다. 입력 a, b, c, d에 대하여 out1은 AND gate, out2는 OR gate 연산을 수행한 결과가 출력된다.

2)

코드

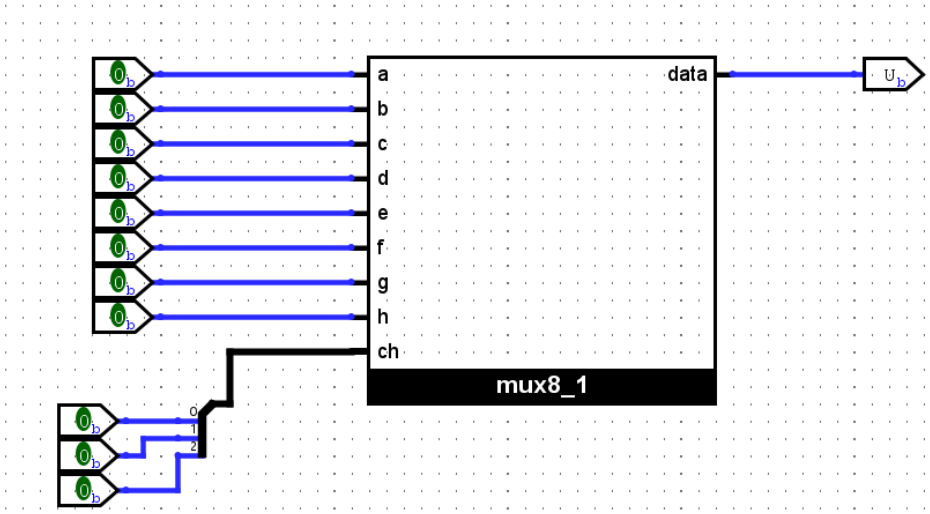
C: > Users > dlrjs > Desktop > 디논설 과제2 > 8to1_multiplexer.vhd

```vhdl
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

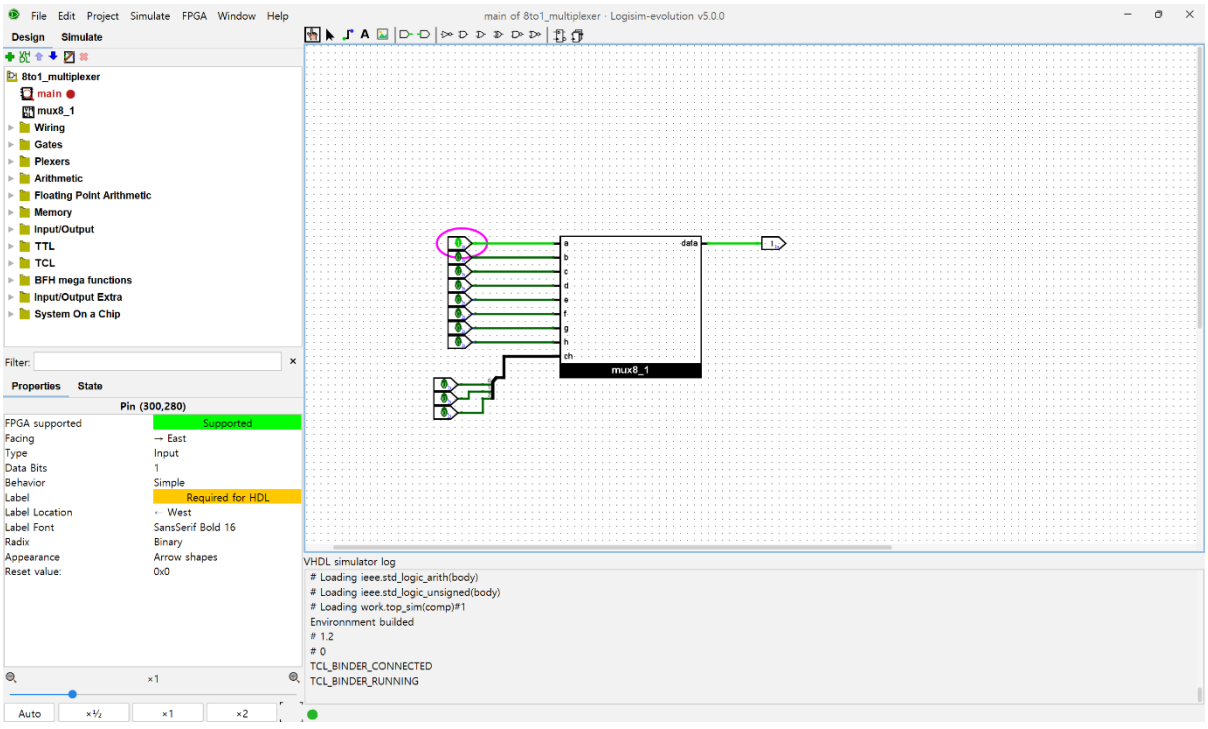
entity mux8_1 is
    port(
        a,b,c,d,e,f,g,h :in std_logic;
        ch              :in std_logic_vector(2 downto 0);
        data            :out std_logic
    );
end entity mux8_1;

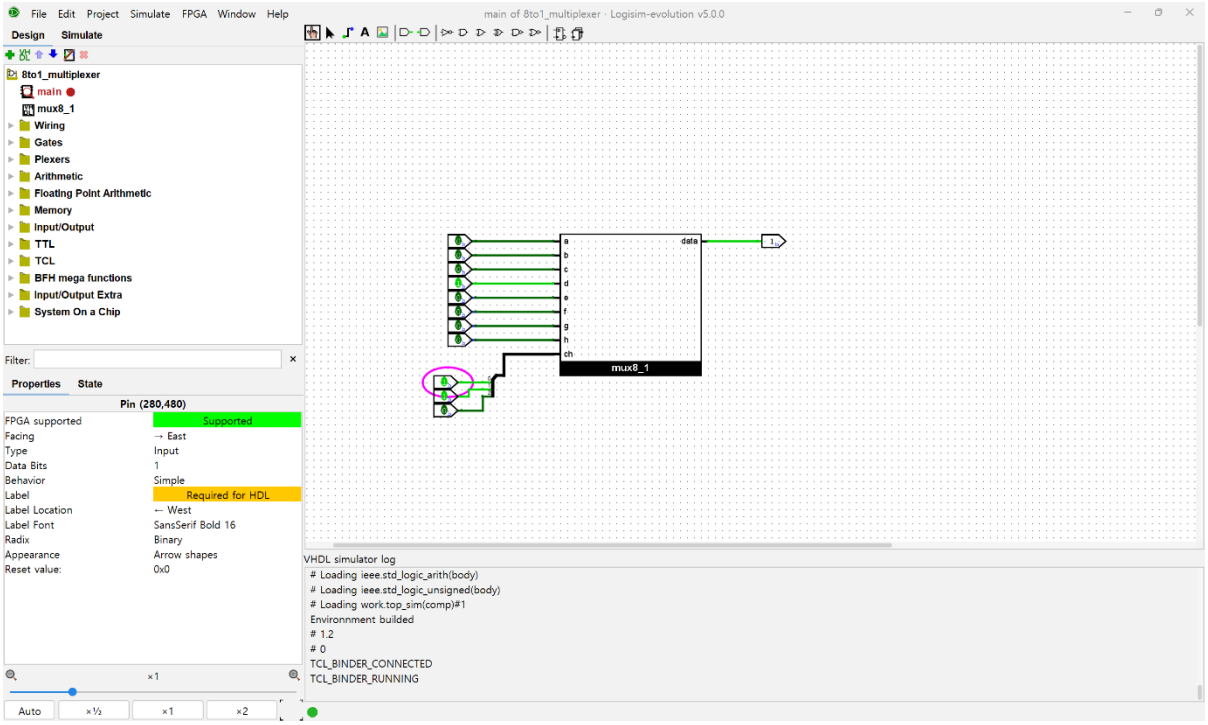
architecture bhv of mux8_1 is
begin
    process(a,b,c,d,e,f,g,h,ch)
    begin
        case ch is
            when "000" => data <= a;
            when "001" => data <= b;
            when "010" => data <= c;
            when "011" => data <= d;
            when "100" => data <= e;
            when "101" => data <= f;
            when "110" => data <= g;
            when "111" => data <= h;
            when others => data <= '0';
        end case;
    end process;
end architecture bhv;
```

회로


a
b
c
d
e
f
g
h
ch
data
mux8_1
0
1
2

시뮬레이션 결과


File Edit Project Simulate FPGA Window Help
main of 8to1_multiplexer · Logisim-evolution v5.0.0
Design Simulate
8to1_multiplexer
main
mux8_1
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (300,280)
FPGA supported Supported
Facing East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING
Auto



8-to-1 multiplexer 회로로 ch에 입력된 값에 따라 출력할 값을 선택할 수 있는 회로이다. a ~ h까지의 입력이 ch의 입력 값에 해당하는 값이 아니면 출력되지 않는다.

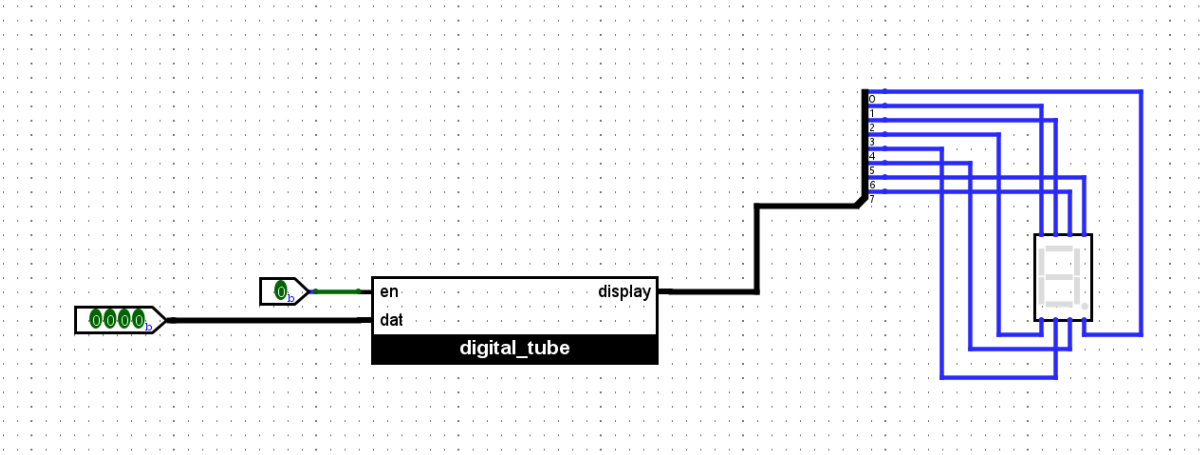
3)

코드

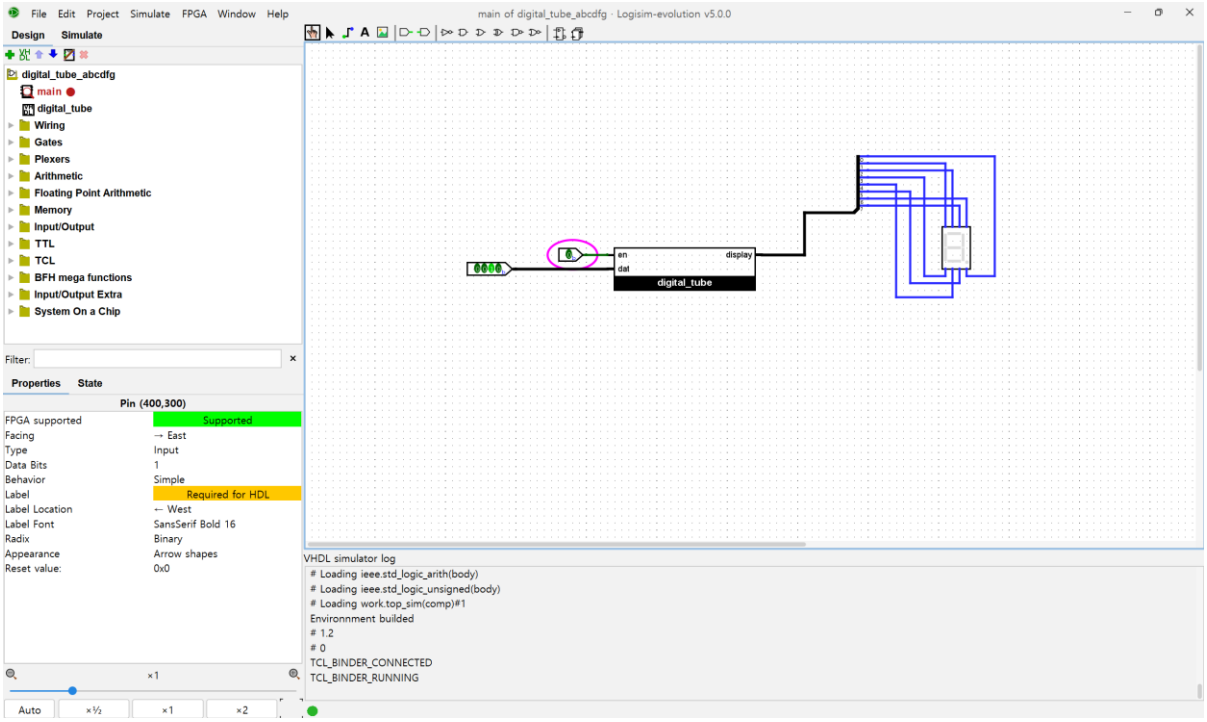
C: > Users > dlrjs > Desktop > 디논설 과제2 > digital_tube.vhd

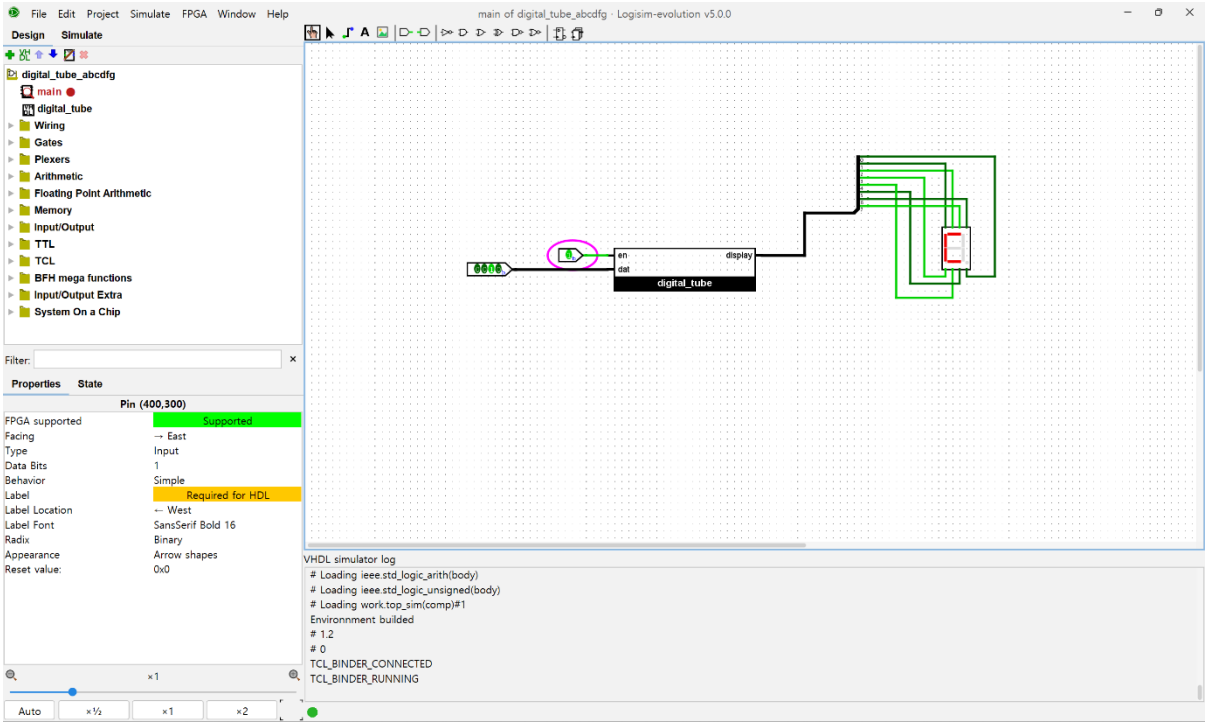
```
1   library ieee;
2   use ieee.std_logic_1164.all;
3   use ieee.std_logic_arith.all;
4   use ieee.std_logic_unsigned.all;
5
6   entity digital_tube is
7       port(
8           en      :in std_logic;
9           dat     :in std_logic_vector(3 downto 0);
10          display :out std_logic_vector(7 downto 0)
11      );
12  end entity digital_tube;
13
14  architecture bhv of digital_tube is
15  begin
16      process(en,dat)
17      begin
18          if en='0' then
19              display <= "ZZZZZZZZ";
20            else
21              case dat is
22                  when "0000" => display <= "11101110";
23                  when "0001" => display <= "00111110";
24                  when "0010" => display <= "10011100";
25                  when "0011" => display <= "01111010";
26                  when "0100" => display <= "10001110";
27                  when "0101" => display <= "11110110";
28                  when others => display <= "00000000";
29              end case;
30          end if;
31      end process;
32  end architecture bhv;
```

회로


0
1
2
3
4
5
6
7
0
0000
en
dat
display
digital_tube

시뮬레이션 결과


File Edit Project Simulate FPGA Window Help
main of digital_tube_abcdfg · Logisim-evolution v5.0.0
Design Simulate
digital_tube_abcdfg
main
digital_tube
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
Pin (400,300)
FPGA supported Supported
Facing → East
Type Input
Data Bits 1
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
×1
Auto ×½ ×1 ×2
en
dat
display
digital_tube
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1,2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING



Digital tube의 일종인 7-segment를 이용한 회로이다. dat에 입력된 값에 따라 해당되는 알파벳(a, b, c, d, f, g)이 출력된다.

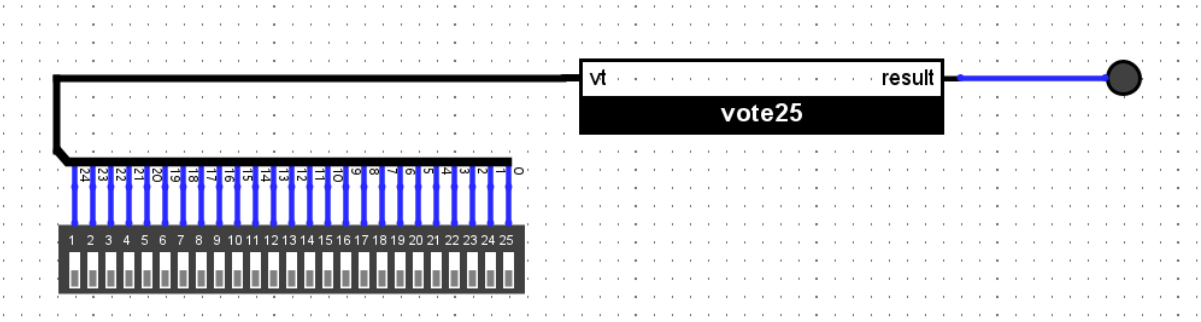
4)

코드

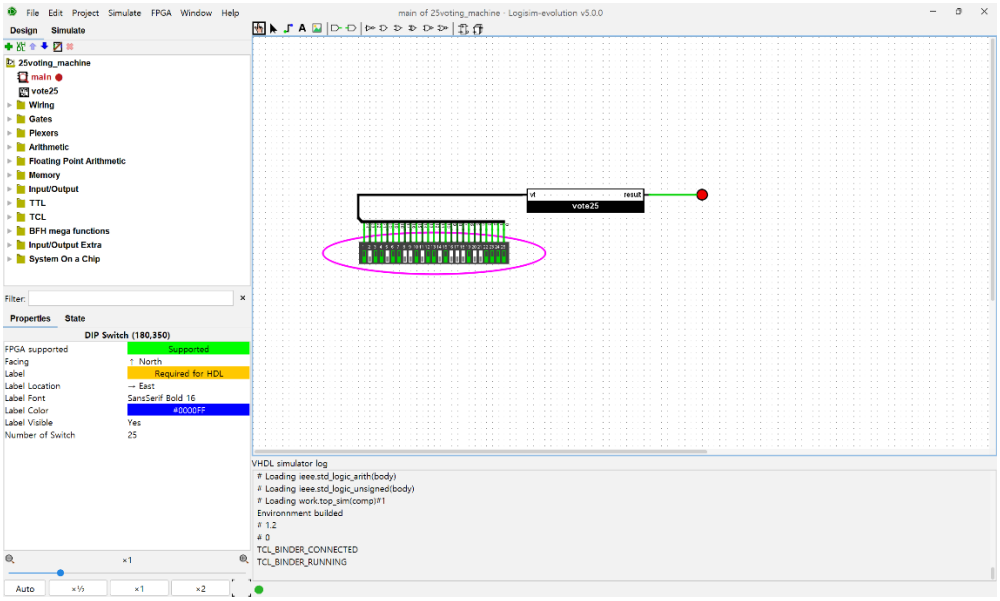
C: > Users > dlrjs > Desktop > 디논설 과제2 > 25_voting_machine.vhd

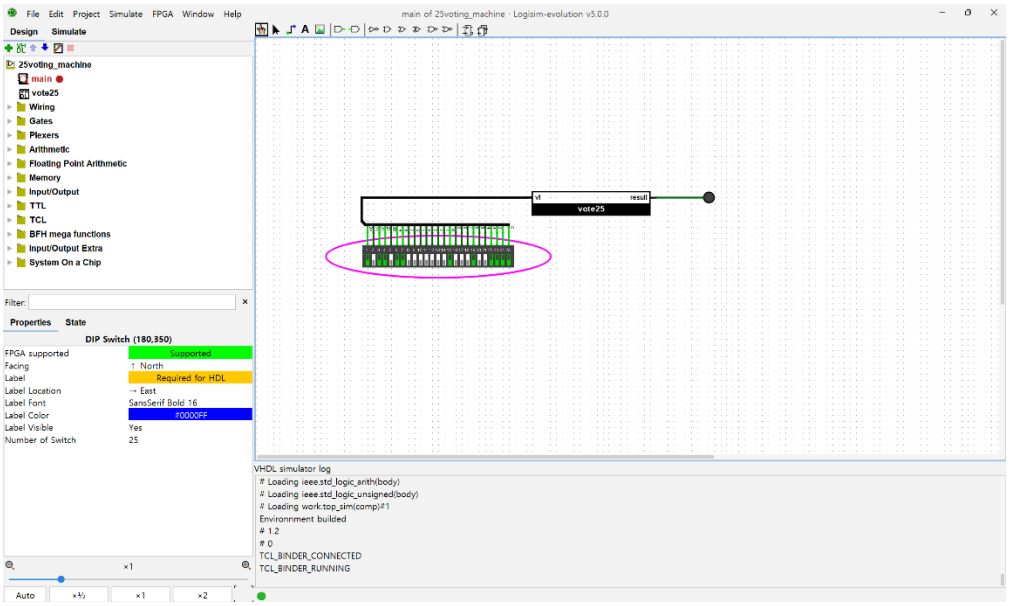
```
1   library ieee;
2   use ieee.std_logic_1164.all;
3   use ieee.std_logic_arith.all;
4   use ieee.std_logic_unsigned.all;
5
6   entity vote25 is
7       port(
8           vt        :in std_logic_vector(25 downto 1);
9           result    :out std_logic
10      );
11  end entity vote25;
12
13  architecture bhv of vote25 is
14  begin
15      process(vt)
16      variable sum:integer range 0 to 25;
17      begin
18          sum:= 0;
19          for i in 1 to 25 loop
20              if (vt(i)='1') then
21              sum:=sum+1;
22                  if sum >= 13 then
23                      result <= '1';
24                  else
25                      result <= '0';
26                  end if;
27              end if;
28          end loop;
29      end process;
30  end architecture bhv;
```

회로


vt
result
vote25
24 23 22 21 20 19 18 17 16 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

시뮬레이션 결과


File Edit Project Simulate FPGA Window Help
main of 25voting_machine · Logisim-evolution v3.0.0
Design Simulate
25voting_machine
main
vote25
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
DIP Switch (180,350)
FPGA supported Supported
Facing ↑ North
Label Required for HDL
Label Location → East
Label Font SansSerif Bold 16
Label Color #0000FF
Label Visible Yes
Number of Switch 25
vt
result
vote25
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING


File Edit Project Simulate FPGA Window Help
main of 25voting_machine · Logisim-evolution v3.0.0
Design Simulate
25voting_machine
main
vote25
Wiring
Gates
Plexers
Arithmetic
Floating Point Arithmetic
Memory
Input/Output
TTL
TCL
BFH mega functions
Input/Output Extra
System On a Chip
Filter:
Properties State
DIP Switch (180,350)
FPGA supported Supported
Facing ↑ North
Label Required for HDL
Label Location → East
Label Font SansSerif Bold 16
Label Color #0000FF
Label Visible Yes
Number of Switch 25
vt
result
vote25
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING

25-person voting machine 회로이다. 25명의 찬반 투표를 집계하여 과반수를 넘으면 led에 불이 들어온다.

5)

코드

C: > Users > dlrjs > Desktop > 디논설 과제2 > p5.vhd

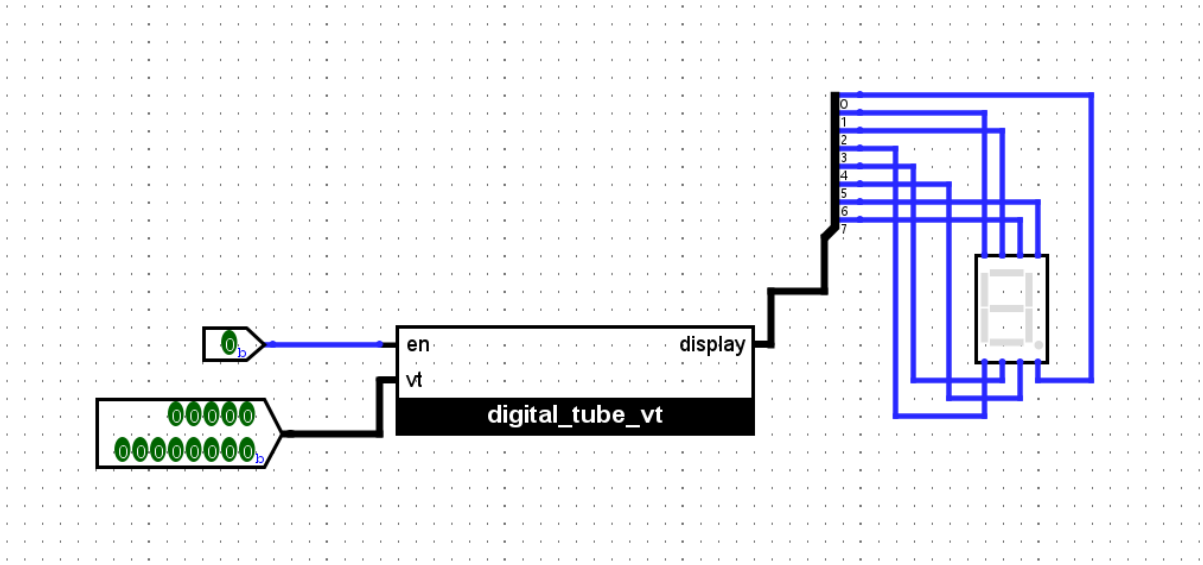
```
library ieee;
use ieee.std_logic_1164.all;
use ieee.std_logic_arith.all;
use ieee.std_logic_unsigned.all;

entity digital_tube_vt is
    port(
        en      :in std_logic;
        vt      :in std_logic_vector(13 downto 1);
        display :out std_logic_vector(7 downto 0)
    );
end entity digital_tube_vt;

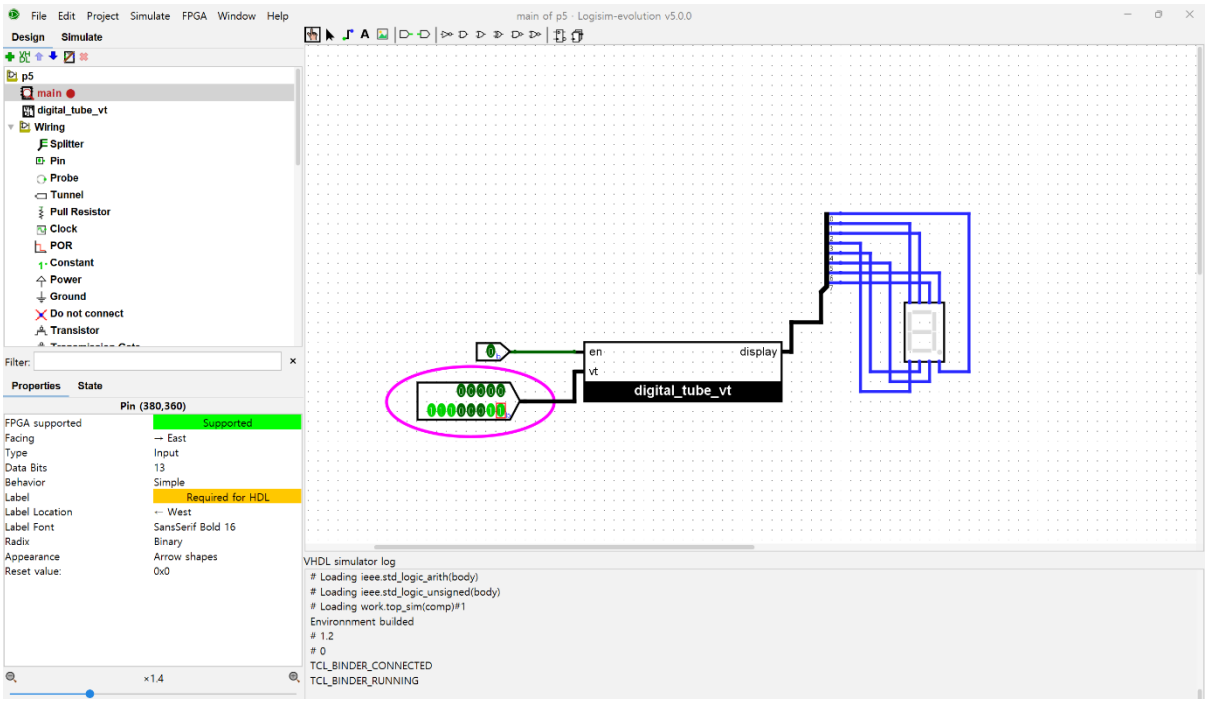
architecture bhv of digital_tube_vt is
begin
    process(en,vt)
    variable sum:integer range 0 to 13;
    begin
        sum:= 0;
        for i in 1 to 13 loop
            if (vt(i)='1') then
                sum:= sum + 1;
            end if;
        end loop;

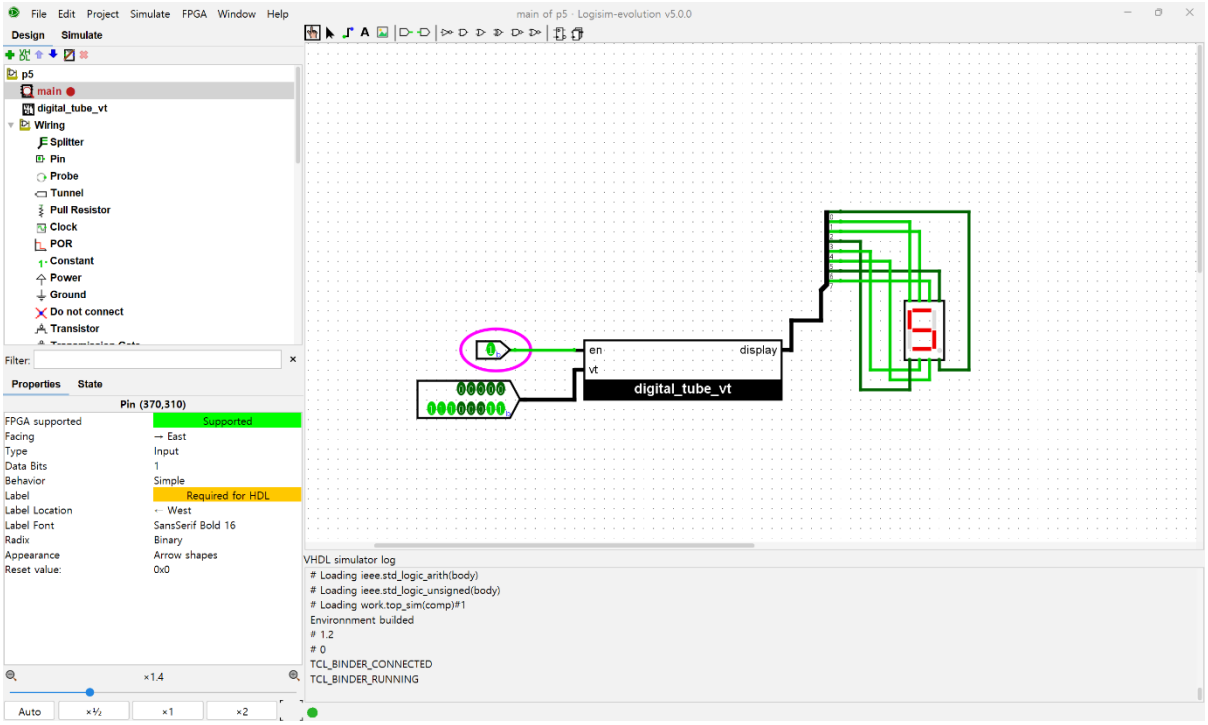
        if en='0' then
            display <= "ZZZZZZZZ";
          else
            case sum is
                when 0 => display <= "11111100";
                when 1 => display <= "01100000";
                when 2 => display <= "11011010";
                when 3 => display <= "11110010";
                when 4 => display <= "01100110";
                when 5 => display <= "10110110";
                when 6 => display <= "10111110";
                when 7 => display <= "11100000";
                when 8 => display <= "11111110";
                when 9 => display <= "11100110";
                when 10 => display <= "11101110";
                when 11 => display <= "00111110";
                when 12 => display <= "10011100";
                when 13 => display <= "01111010";
                when others => display <= "00000000";
            end case;
        end if;
    end process;
end architecture bhv;
```

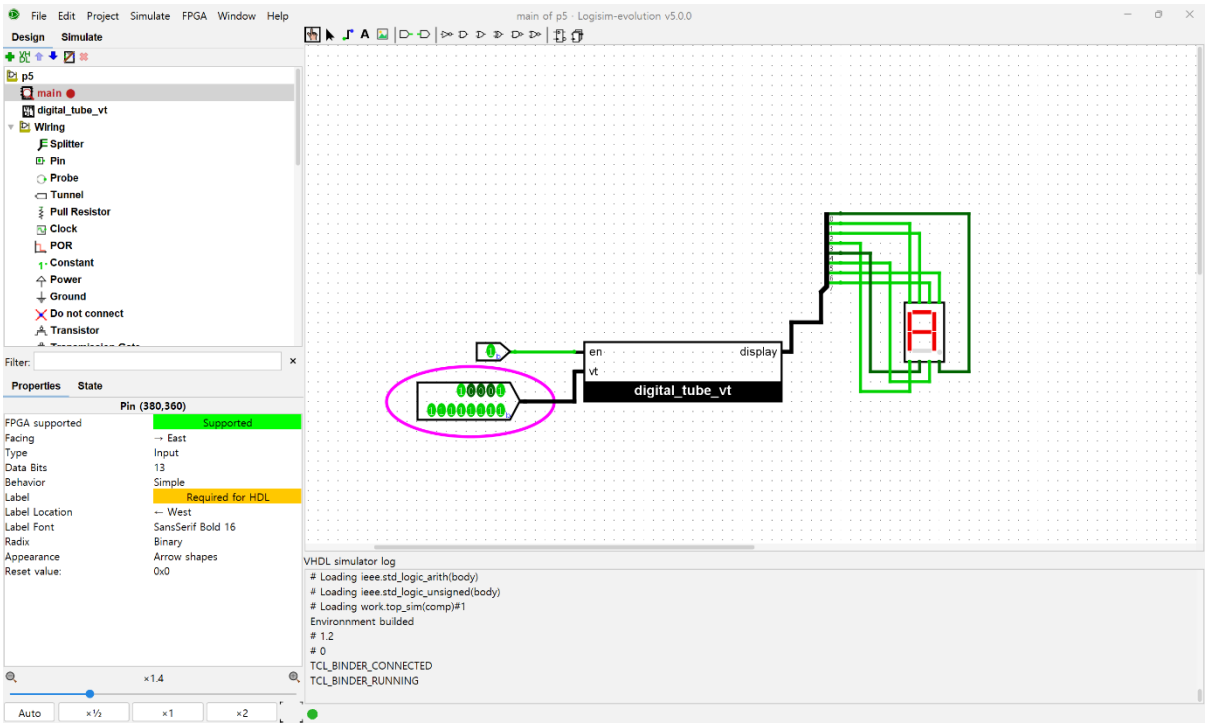
회로


0
1
2
3
4
5
6
7
0
en
vt
display
digital_tube_vt
00000
00000000

시뮬레이션 결과


File Edit Project Simulate FPGA Window Help
main of p5 - Logisim-evolution v5.0.0
Design Simulate
p5
main
digital_tube_vt
Wiring
Splitter
Pin
Probe
Tunnel
Pull Resistor
Clock
POR
Constant
Power
Ground
Do not connect
Transistor
Filter:
Properties State
Pin (380,360)
FPGA supported Supported
Facing → East
Type Input
Data Bits 13
Behavior Simple
Label Required for HDL
Label Location ← West
Label Font SansSerif Bold 16
Radix Binary
Appearance Arrow shapes
Reset value: 0x0
×1.4
en
vt
display
digital_tube_vt
VHDL simulator log
Loading ieee.std_logic_arith(body)
Loading ieee.std_logic_unsigned(body)
Loading work.top_sim(comp)#1
Environnment builded
1.2
0
TCL_BINDER_CONNECTED
TCL_BINDER_RUNNING





en에 1이 입력됐을 때, vt에 입력된 1의 개수에 따라 해당되는 수(0~9)와 알파벳(a~f)을 7-segment에 출력하는 회로이다. vt가 13bit vector이므로 최대 알바펫 d까지 출력된다. en에 0이 입력된 경우, 어떤 값도 출력하지 않는다.